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Correlating paleomagnetic, geochemical and petrographic evidence to date diagenetic and fluid flow events in the Mississippian Turner Valley Formation, Moose Field, Alberta, Canada

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Abstract

Petrographic, geochemical and paleomagnetic analyses of the Mississippian Turner Valley Formation provide constraints on diagenesis and fluid flow events in the Western Canada Sedimentary Basin. Paleomagnetic plugs and companion geochemical samples were taken from two drillcores, with Fullbore MicroImage log orientations. Dolomite from both wells yielded two magnetization directions. The low-temperature, low-coercivity direction is a drilling-induced remanence rather than a viscous remanent magnetization. The high-temperature, high-coercivity remanence direction is Cretaceous, and there is no sign of a primary Mississippian direction. Geochemical analyses of matrix dolomite yield $\delta^{18}O$ values ranging from 0.65 to −3.34‰ (VPDB standard) and $\delta^{13}C$ values ranging from 1.77 to 3.05‰ VPDB. The least depleted samples have stable isotope values consistent with, or only slightly depleted from, postulated Mississippian dolomite values. The remaining sample values exhibit a negative covariant trend consistent with either mixing with another diagenetic fluid or recrystallization during burial. Petrographic analysis reveals the presence of a recrystallization event that caused zoning and a gradual increase size of the dolomite crystals. This event is thought to have caused both the Cretaceous paleomagnetic remanence and the altered geochemical values. The minor enrichment in Sr radiogenic isotopes, relative to coeval seawater values, suggests that both an extrabasinal source for any fluid and large-scale fluid flow are unlikely. The results also indicate that magnetic remanences are very sensitive to visually minor changes in carbonate recrystallization from heat or pressure, so that great care must be taken in correlating paleomagnetic and geochemical data. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; stable isotope geochemistry; carbonates; diagenesis; Western Canada Sedimentary Basin; Mississippian

1. Introduction

Previous work has suggested close links between orogenically derived fluid flow and magnetization ages observed in the associated foreland. For example, many Mississippi Valley type lead–zinc deposits show a close correlation between their paleomagnetically determined age and major tectonic events (see Symons et al., 1996, for discussion). A second example is the Permian ‘Kiaman overprint’. This widespread secondary magnetization is found in

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most Paleozoic carbonate and clastic formations in eastern North America, and its timing correlates with that of the Alleghanian and Ouachita orogenies (Oliver, 1986, 1992; McCabe and Elmore, 1989; Bethke and Marshak, 1990). Precipitation of magnetic minerals from large-scale fluid migration driven by sediment compaction, tectonic 'squeezing' or gravity, is usually invoked as the causal mechanism for the magnetization (Jackson and Beales, 1967; Cathles and Smith, 1983; Garven, 1985; Bethke, 1986). Thus, paleomagnetism can be used to date fluid migration, provided the fluid is capable of forming remanence-carrying minerals. An extension of this hypothesis suggests that, if an area has undergone two fluid flow events, it could then be possible to date both events (Lewchuk and Symons, 1996). For calcareous sedimentary rocks, the observed paragenetic sequence sets out the relative timing of diagenetic events, and paleomagnetic techniques can date events when magnetic minerals are precipitated or recrystallized.

The first purpose of this study is an investigation of drill core orientation using paleomagnetic techniques. Modern techniques, such as Fullbore MicroImage logging, enable one to correlate major structures in the wall rocks to structures within the core, and thereby to orient the core. However, such logs are unavailable for most older wells, in which case paleomagnetism often provides an alternative method of orientation. In many core samples, low coercivity/low temperature or viscous remanent magnetization (VRM) is present that is aligned to either the present Earth's magnetic field (PEMF) or the dipole-averaged Earth's magnetic field (DEMF) (Kodama, 1984; Rolph et al., 1995; Lewchuk et al., 1998). By measuring the orientation of its VRM, drillcore can be absolutely oriented in 3-D space so that directional properties such as fracture orientations can be obtained for reservoir analysis. The second purpose of this study is to use paleomagnetic, geochemical and petrographic techniques in combination to study a Mississippian carbonate gas reservoir in the Western Canada Sedimentary Basin (WCSB) (Fig. 1). Correlation of the magnetization components to the paragenetic sequence, determined through petrographic and geochemical techniques, allows us to date major diagenetic events such as dolomitization-inducing and/or hydrocarbon-bearing fluid migration. This is important in the WCSB where the timing of both reservoir-creating dolomitization and of hydrocarbon-bearing fluid migration has been the subject of debate for a number of years (Nesbitt and Muehlenbachs, 1994; Qing and Mountjoy, 1994).

2. Regional geologic setting and stratigraphy

Eastward or cratonic-verging thrusting occurred throughout eastern British Columbia and western Alberta during the Jurassic/Cretaceous Columbian and Laramide orogenies, creating the thrusts and folds of the Foreland Belt of the Canadian Cordillera. Moose Mountain formed in a series of overlapping thrust sheets that are underlain by low-angle sole faults (Fig. 2). The surface sheet is folded into an anticlinal structure because its sole fault steps up from the Cambrian to Devonian strata. At the surface this anticlinal structure forms Moose Mountain, an elongated dome with its long axis oriented northwest–southeast (Ower, 1975). The dome is bounded to the east by the Moose Mountain sole fault and to the west by the Prairie Mountain and Forgetmenot thrust faults. Mississippian strata of the Banff Formation are exposed at the center of the dome.

The Mississippian Turner Valley Formation lies within the platform carbonates of the Rundle assemblage in the Foothills of southern Alberta (Fig. 1). In the Moose Mountain area, 80 km southwest of Calgary, the Turner Valley Formation is overlain by the Mount Head Formation and underlain by the Shunda and Pekisko formations (Fig. 2). At the surface, the Turner Valley Formation consists of enchinoderm-rich limestones and dolomites (Macqueen and Bamber, 1967). In the subsurface, the formation consists of three informal members: the 'lower porous', 'middle dense', and 'upper porous' members. The lower and upper members consist of echinoderm and echinoderm-bryozoan grainstones and packstones, whereas the middle member consists of mostly dolomitized wackestones and mudstones (Al-Aasm and Lu, 1994).

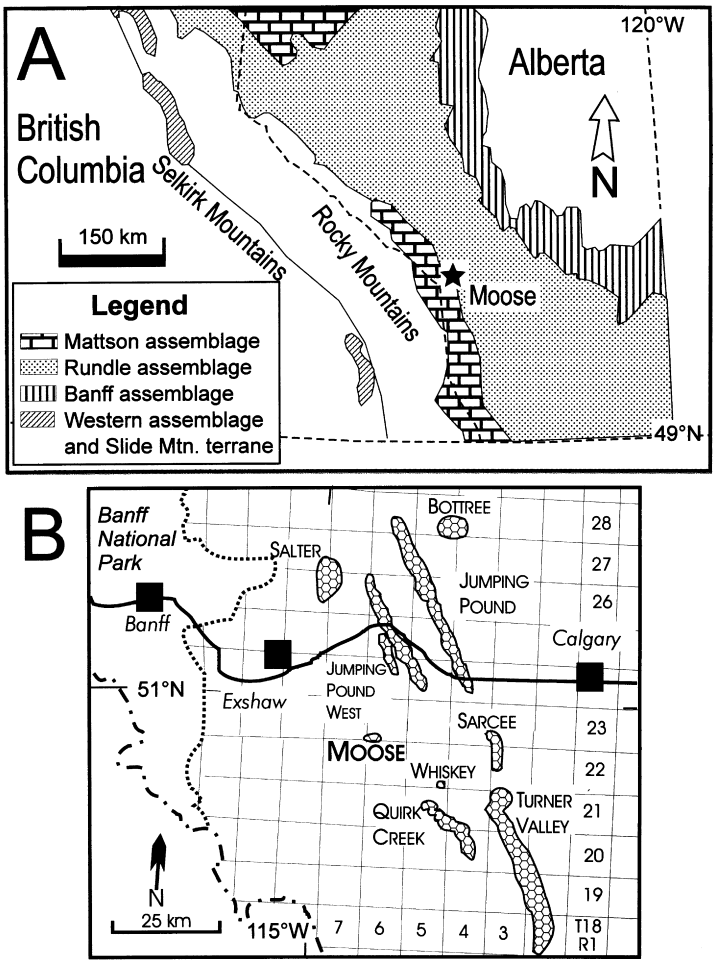


Fig. 1. (a) Mississippian strata in the WCSB and the location of the Moose field. (b) Moose field and surrounding Mississippian reservoirs.

3. Methodology

Core from wells 10–14–23–7W5 or Well 14 and 2–23–23–7W5 or Well 23 in the Mississippian Turner Valley Formation of the Moose Field was sampled for analysis at the Alberta Energy Utilities Board core facility in Calgary.

3.1. Petrography and geochemistry

The petrographic and geochemical samples were collected from as close to the paleomagnetic plug samples as possible to directly compare their data to the petrographic and geochemical data. Twenty-five thin sections were examined using a standard petrographic microscope and a microscope with a Technosyn cold cathodoluminescence stage operating at a current voltage of 12–15 kV and intensity of ~420 μA. Additionally, fluorescence characteristics were examined with a Nikon EPI fluorescence stage connected to a petrographic microscope.

Oxygen and carbon isotopes were extracted from dolomite and calcite samples ($n = 22$) using a microscope-mounted drill assembly. The samples were reacted in vacuo with 100% pure phosphoric acid for at least 4 h at either 25°C for calcite or 50°C for dolomite using the method described by Al-Aasm et al. (1990). The evolved CO_2 gas was analyzed for isotopic ratios on an SIRA-12 mass spectrometer. The values of O and C isotopes are reported in per

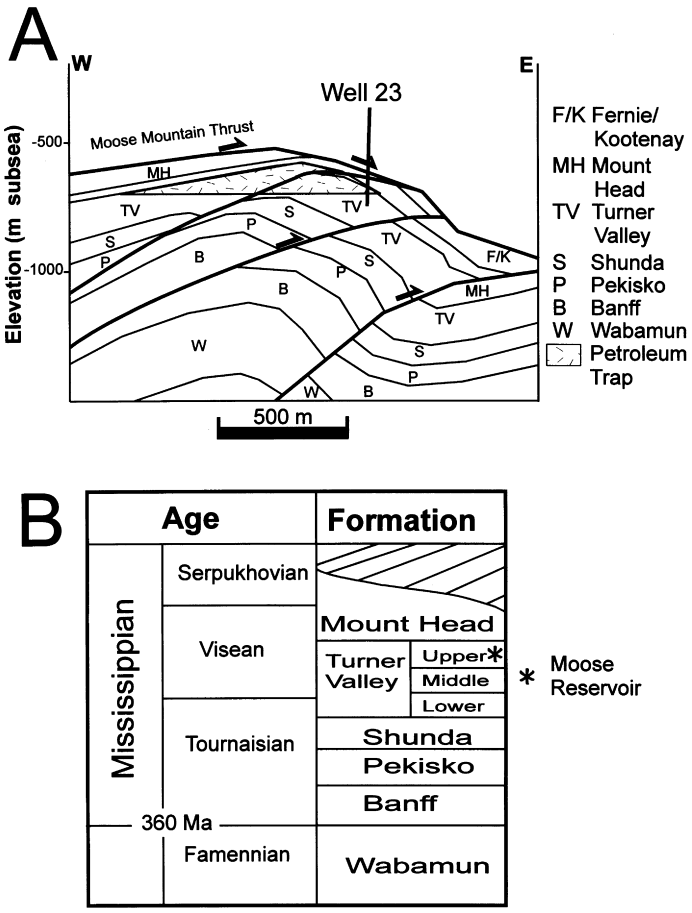


Fig. 2. (a) Cross-section of the Moose field below the Moose Mountain Thrust Fault, adapted from D.A. Keiver (pers. comm. to K.P. Gillen, 1998). (b) Stratigraphic position of the Turner Valley Formation.

mil (‰) relative to the VPDB standard and were corrected for phosphoric acid fractionation. Precision was better than 0.05‰ for both $\delta^{18}O$ and $\delta^{13}C$. Strontium isotopes were analyzed for selected dolomite, calcite and anhydrite samples ($n = 6$). Sr isotopic ratios were measured on a Finnigan MAT 261 mass-spectrometer equipped with nine collectors. All analyses were performed in the static multi-collector mode using Re filaments. Correction for isotope fractionation during the analyses was made by normalization to $^{86}Sr/^{88}Sr = 0.1194$. The mean standard error of mass spectrometer performance was 0.00003 for NBS-987.

Major and minor element analyses (Ca, Mg, Fe, Sr, and Mn) were carried out on highly polished carbon-coated thin sections with a Cameca Camebax SX50 electron microbeam analyzer equipped with three crystal spectrometers and a back-scatter electron detector. The operating conditions were an accelerating voltage of 15 kV, a measured beam current of 6 nA, and a beam diameter of 4–10 μm. The standards used were wollastonite (Ca), MgO (Mg), $MnTiO_3$ (Mn), hematite (Fe), and strontianite (Sr). Precision was better than 0.5 mol% for all elements. Due to the low concentrations of Mn, Fe and Sr, the EMP data are considered here to be semiquantative. Nevertheless, efforts were made to have a detection limit better than about 50 ppm by performing the analyses on the basis of the strongest wave-lengths of these elements and by analyzing each element separately.

3.2. Paleomagnetism

Each core was re-assembled into segments that were as long as possible on which an arbitrary master orientation line (MOL) was drawn to provide a reference position for the paleomagnetic plugs. The 2.5 cm diameter plugs were drilled perpendicular to the core axis, and their azimuths were measured relative to the MOL. Each plug was sliced into one to three paleomagnetic specimens of ~2.2 cm length. The potential for introducing spurious magnetizations during well drilling has been recognized (Pinto and McWilliams, 1990) and therefore, the plugs were taken as far from the outer edges of the core as possible. Fullbore MicroImage logs were used to determine the fracture and bedding orientations of segments relative to both their MOL and the surrounding bedrock, so that the true orientation of their MOL was determined. Rotation of magnetization directions along with rotation of the MOL to its true orientation, including correction for the borehole deviation from vertical, resulted in the orientation of the specimens in 3-D space, so that their true paleomagnetic directions could be determined.

Prior to analysis, the specimens were stored for three months in a magnetically shielded room with a field intensity of $\ll 1\%$ of the Earth's magnetic field in order to allow the least-stable laboratory magnetization components to decay. All subsequent testing on the specimens was done in the shielded room, minimizing the possibility of the acquisition of a spurious magnetization. The natural remanence magnetization (NRM) of each specimen was measured using a Canadian Thin Films CTF-420 cryogenic magnetometer which can detect a magnetization intensity down to about 5×10^{-6} A/m for a standard paleomagnetic specimen. If two specimens were obtained from a plug, one was subjected to alternating field (AF) demagnetization in about 25 steps from 2 to 140 mT using a Sapphire Instruments SI-4 AF demagnetizer and the other was thermally demagnetized in about 30 steps from 80 to 580°C using a Magnetic Measurements MMTD-1 thermal demagnetizer. If only a single specimen was obtained, it was thermally demagnetized, and for plugs with three specimens, two were thermally and one was AF demagnetized. Thermal demagnetization was used preferentially because previous work in the WCSB had suggested that it was more effective for isolating the magnetic components (Lewchuck et al., 1998).

The individual remanence vector components from the specimens were obtained using the principal component method of Kirshvinck (1980). Most components were defined over at least three demagnetization steps and all had maximum angular deviation angles of $<13°$. Component mean directions for groups of specimens were calculated using Fisher (1953) statistics.

Rock magnetic experiments were done to determine the mineralogy and grain size of the main magnetic minerals. Isothermal remanent magnetization (IRM) acquisition paths were measured by progressively magnetizing the specimen in 15 steps with an increasing direct current (DC) field up to a maximum of 900 mT using a Sapphire Instruments SI-6 pulse magnetizer to reach the saturation isothermal remanent magnetization (SIRM) and then AF demagnetizing the SIRM in 10 steps up to 100 mT.

4. Results

4.1. Petrography and geochemistry

The diagenetic history of the Moose field includes episodes of early silicification, matrix-destructive dolomite formation, brecciation, anhydritization, compaction, and late fluorite and calcite vein-filling (Fig. 3). The paragenetic sequence is based on petrographic observations and corroborated with geochemical evidence. The most significant and relevant diagenetic events are discussed below.

4.1.1. Dolomitization

Four types of dolomite are present in the Turner Valley Formation. From oldest to youngest they are: pervasive matrix microdolomite, pseudomorphic dolomite, mesodolomite, and vug-rimming coarse dolomite.

The pervasive microdolomite is a fabric-destructive, non-ferroan, subhedral dolomite that replaces the matrix and allochems (Figs. 4a and 5a). The crystals are typically 5–30 μm in size and non-stoichiometric (mean $CaCO_3$ 56.1 mol%; Table 1). In some samples, this dolomite also occurs in

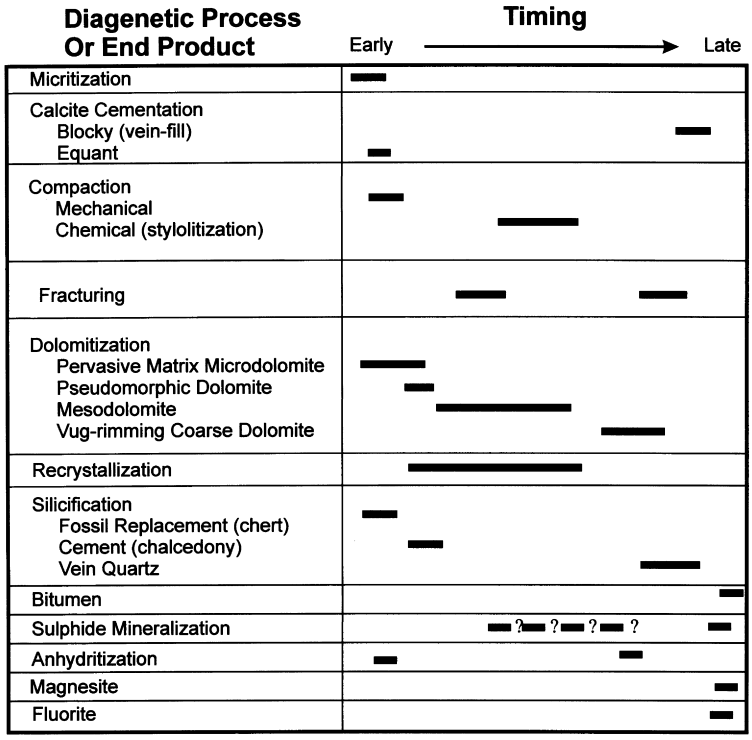


Fig. 3. Paragenetic sequence of the Turner Valley carbonates in the Moose field.

association with algal filaments and nodular anhydrite. This dolomite has the heaviest δ^{18} O values of all the dolomites (+0.66 to −3.34‰ VPDB; Table 2; Fig. 6) but these values are still slightly depleted relative to postulated Mississipian marine dolomite (Al-Aasm and Lu, 1994; Durocher and Al-Aasm, 1997). The pervasive microdolomite δ^{13}C values also fall close to those postulated for Mississippian marine dolomite (Fig. 6). $^{87}Sr/^{86}Sr$ values of this dolomite are close to Mississippian seawater composition (cf. Denison et al., 1994).

Mesodolomite crystals range from 40–100 μm (Fig. 5c) and show a gradual increase in size from the microdolomite range to the largest mesodolomite crystal size. Mesodolomite is anhedral to subhedral, non-ferroan, non-stoichiometric (mean $CaCO_3$ 56.1 mol%; Table 1) and sometimes shows concentric zoning with dark cores and bright rims under UV illumination, although the contact between the core and the rim is irregular. The mesodolomite apparently postdates microdolomite as suggested by its larger crystal size. It shows a covariant trend in its isotopic composition, unlike that of microdolomite (Fig. 6) with its δ^{18}O values ranging from −3.74 to −7.51‰ VPDB, while its δ^{13}C values range from +0.65 to +2.54‰ VPDB (Table 2), and the correlation coefficient (R^2) for the six samples is 0.79. However, the mesodolomite's $^{87}Sr/^{86}Sr$ values are slightly more radiogenic than the Mississippian seawater composition (Fig. 7).

Pseudomorphic dolomite (e.g. Sibley, 1982) forms as a single, non-ferroan, inclusion-rich crystal of dolomite that mimics the morphology of crinoid ossicles. Based on its relationship with mesodolomite, it appears that the formation of this dolomite overlaps with the mesodolomite or formed slightly earlier.

Vug-rimming coarse dolomite cement occurs in some samples, postdating the earlier dolomites. It is euhedral, 100 μm in size, non-ferroan and non-stoichiometric (mean $CaCO_3$ 55.3 mol%; Table 1). Under UV illumination, this dolomite is occasionally zoned. Bitumen is often found lining the coarse dolomite in the vugs, suggesting that bitumen formation postdates the dolomite.

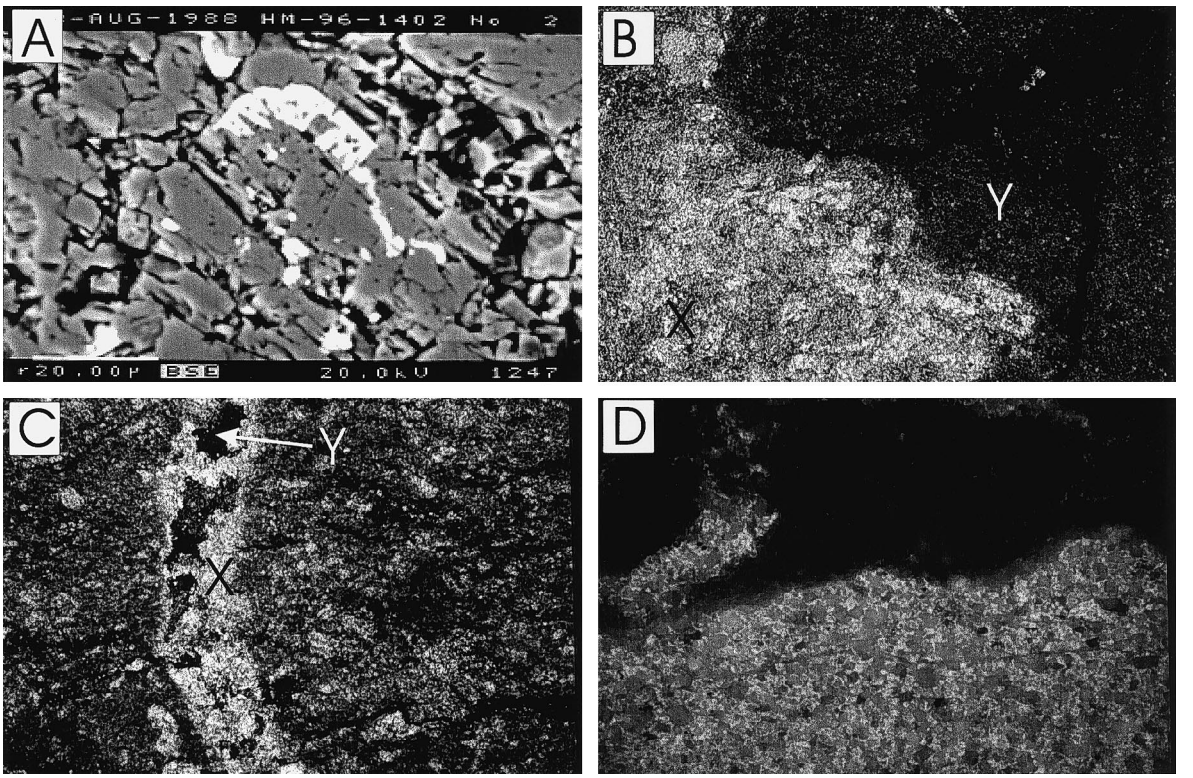


Fig. 4. Photomicrographs: (a) backscatter SEM image of replacement of calcite (white) by dolomite (dark grey); (b) plane light photomicrograph of nodular anhydrite (X) and early matrix microdolomite (Y); (c) polarized light photomicrograph of vein-filling calcite (X) and fluorite (Y) in matrix dolomite; and (d) ultraviolet light photomicrograph of dolomite (light) and organics (dark). The horizontal fields of view for (b), (c), and (d) are 2.1, 1.0, and 1.0 mm, respectively.

4.1.2. Calcite cementation

Calcite cementation is a very minor process in the diagenetic history of the Turner Valley Formation in the Moose Field. Early-precipitated equant calcite cement occupies pore spaces in crinoidal grainstones and coated grains. It has a low-Mg concentration (mean $MgCO_3$ 0.08 mol%), undetectable Mn and Fe, and a high Sr concentration (mean = 988 ppm; Table 1). Later vein-fill blocky calcite crosscuts the dolomite phases (Fig. 4c). The carbon and oxygen isotopic signatures from one sample of the early calcite cement indicate partial preservation of primary marine isotopic signatures when compared to Mississippian carbonates (Fig. 6), whereas the later calcite cement infilling veins show more depleted oxygen and carbon isotope values. The $^{87}Sr/^{86}Sr$ analysis value from one sample of blocky calcite is more radiogenic than Mississippian seawater (Fig. 7).

4.1.3. Anhydritization

Early, nodular anhydrite is observed in some facies associated with algal filaments and the dolomitized matrix (Fig. 4b). The nodules are subrounded, and usually less than 1 cm in diameter. This anhydrite resembles anhydrite formed in a sabhka environment (e.g. McKenzie, 1981; Butler et al., 1982), but it is a volumetrically minor component of the cores. Secondary anhydrite is found in some fractures and is associated with late fluorite, quartz, magnesite, minor sulfides, and calcite.

4.1.4. Silicification

Silica is a minor rock-forming component. It has a restricted distribution, appearing as coexisting microcrystalline chert, chalcedony and coarser crystalline quartz (Fig. 5b). The silica replaces the original limestone, preserving some primary features (Fig. 5b). In

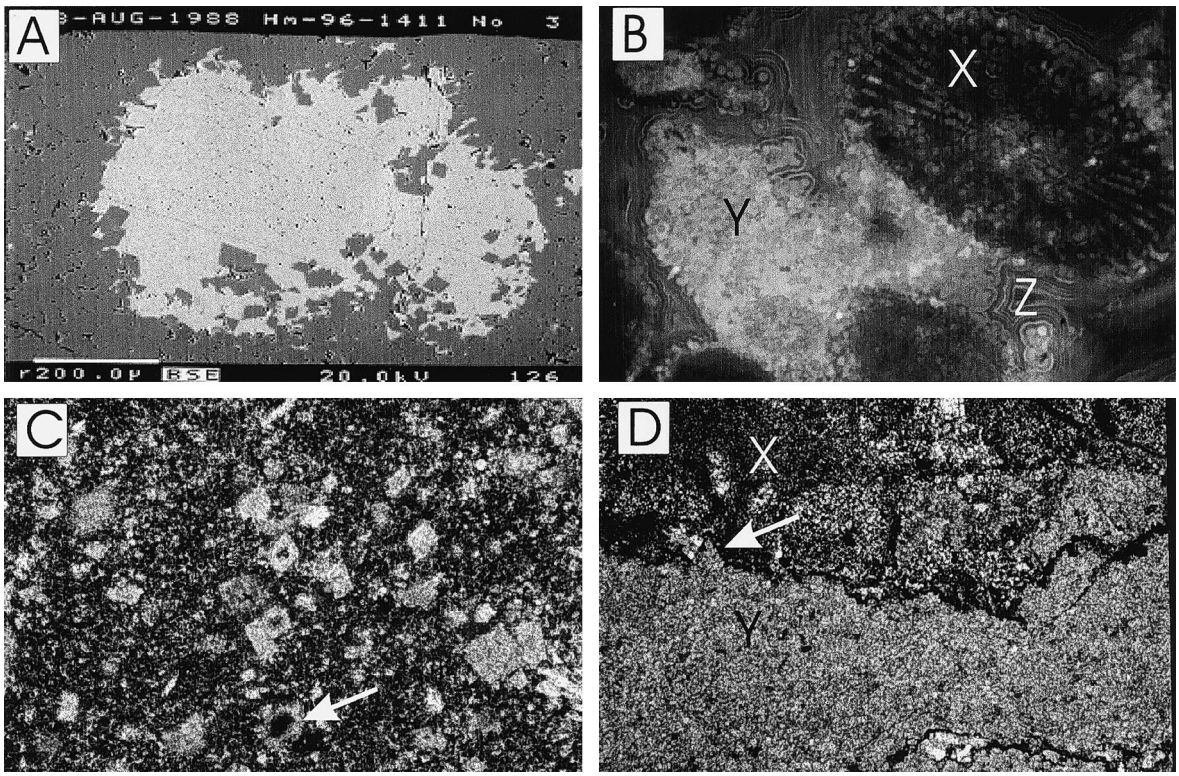


Fig. 5. Photomicrographs: (a) backscatter SEM image of replacement of calcite (white) by euhedral dolomite crystals (dark grey); (b) ultraviolet light photomicrograph of silica components, showing preservation of fossil coral (X), quartz cement infilling (Y), and chalcedony cement (Z); (c) polarized light photomicrograph of bitumen replacement of large mesodolomite crystal (white arrow) in fine-grained matrix dolomite; and (d) polarized light photomicrograph of stylolitic contact (arrow) between massive chert at the top (X), and fine-grained dolomite below (Y). The horizontal fields of view for (b), (c), and (d) are 1.0, 0.5, and 2.1 mm, respectively.

one silicified area, several generations of quartz are present: microcrystalline chert replacing fossils, chalcedonic cement surrounding fossils, coarser cement between the individual fossils, and late vein cements. The fossils are intact, with no mechanical or chemical compaction features apparent (Fig. 5b). Where chert occurs with microdolomite, the contact is either stylolitic (Fig. 5d) or gradational with microdolomite crystals floating in the chert. From these cross-cutting relationships, it appears that silica replaced the fossil during the shallow burial stage and overlapped dolomitization. In contrast, quartz cement infilling some veins is a late diagenetic event.

4.1.5. Late diagenetic phases

Veinlets of fluorite, quartz and ferroan calcite cross-cut all former stages, and must represent late events. These veins are generally <1 mm wide and occur throughout most of the sampled core length (Fig. 4c) except for a larger fluorite vein that is associated with the late quartz. Minor pyrite mineralization occurred after the formation of late anhydrite and appears to have replaced chert and dolomite. It is worthwhile to note here that the sulfides are volumetrically very minor, have very limited distribution, and have a large grain size. Magnesite was also noticed in some samples, where it is associated with the fluorite, quartz and ferroan calcite. Finally, bitumen occurs as a coating on the vug-filling coarse dolomite and other mineral phases and occupying the core of mesodolomite (Fig. 5c).

4.2. Magnetic results

4.2.1. Magnetic mineralogy

The combination of thermal demagnetization, AF

Table 1
Microprobe data for moose field dolomites and calcites

Element	Microdolomite Range Mean No. of samples	Mesodolomite Range Mean No of. Samples	Coarse dolomite Range Mean No of. samples	Calcite cement Range Mean No of. samples
$CaCO_3$ (normalized mol%)	55.2–57.4 56.1 12	55.5–56.6 56.2 7	55.3 – 1	99.6–99.9 99.7 3
$MgCO_3$ (normalized mol%)	44.4–44.7 43.8 12	43.2–44.5 43.7 7	– 44.6 1	0.0–0.2 0.1 3
Mn (ppm)	b.d. – 12	b.d. b.d. 7	b.d. b.d. 1	b.d. b.d. 3
Fe (ppm)	b.d.– 2916 – 12	b.d.– 957 270 7	b.d. b.d. 1	b.d. b.d. 3
Sr (ppm)	b.d.– 471 – 12	b.d.– 373 – 7	205 – 1	449–1819 988 3

b.d. indicates below detection limit; – indicates the calculation was not possible.

demagnetization, IRM acquisition, SIRM demagnetization and backfield analysis data can be used to constrain the magnetic mineralogy and grain size in the specimens. Demagnetization commonly reduced the remanence intensity to <5% of the NRM value or to a value below the magnetometer's sensitivity limit by 450°C or 100 mT, indicating that high-coercivity minerals such as goethite and hematite are not present. The IRM acquisition and demagnetization data, when plotted on magnetite and pyrrhotite SIRM crossover plots (Fig. 8) are most consistent with pseudosingle-domain pyrrhotite which generally has a size of 5–75 μm. Thermally demagnetized specimens do not usually show significant intensity drops between 275 and 325°C, as would be expected if pyrrhotite was the sole remanence carrier. However, minor drops are observed, consistent with a magnetite–pyrrhotite mineral mixture (Dekkers, 1989).

Table 2
Oxygen and carbon isotopic values of the samples studied

Depth	Description	$\delta^{18}O$ (‰ SMOW)	$\delta^{18}O$ (‰ VPDB)	$\delta^{13}C$ (‰ VPDB)
2607.9	MD	29.44	−1.42	2.26
2609.5	MD	31.59	0.66	1.77
2612.1	FD (P)	27.13	−3.67	2.52
2612.1	MD	27.91	−2.92	3.03
2616.8	FD–C/MO	26.09	−4.68	1.91
2616.8	MO	27.06	−3.74	2.54
2617.6	MD	28.21	−2.62	2.76
2620.5	MD	28.14	−2.69	2.36
2621.7	MD	27.46	−3.34	2.18
2623.2	MO	25.87	−4.89	0.94
2623.7	MO	24.55	−6.17	0.65
2624.5	MO	23.17	−7.51	0.85
2625.1	MO	25.06	−5.67	1.44
2627.8	CC	26.29	−4.48	1.13
2627.8	CC–V	22.79	−7.88	−3.10
2627.8	MD	27.62	−3.19	2.13

C—cement; CC—calcite; FD—fossil dolomite; MD—matrix dolomite; MO—mesodolomite; P—porous; and V—vein.

Noting the covariant trend in $\delta^{18}O$ and $\delta^{13}C$, small (<2 cm^3) monomineralic cylinders of dolomite were collected from samples showing the covariant trend, and subjected to IRM acquisition and demagnetization analysis. The analyses for all samples fell into the pseudosingle domain range of pyrrhotite and magnetite (Fig. 9), with the samples having the more negative $\delta^{18}O$ and $\delta^{13}C$ values also having a larger effective magnetic grain size, i.e. falling closer to the multidomain boundary.

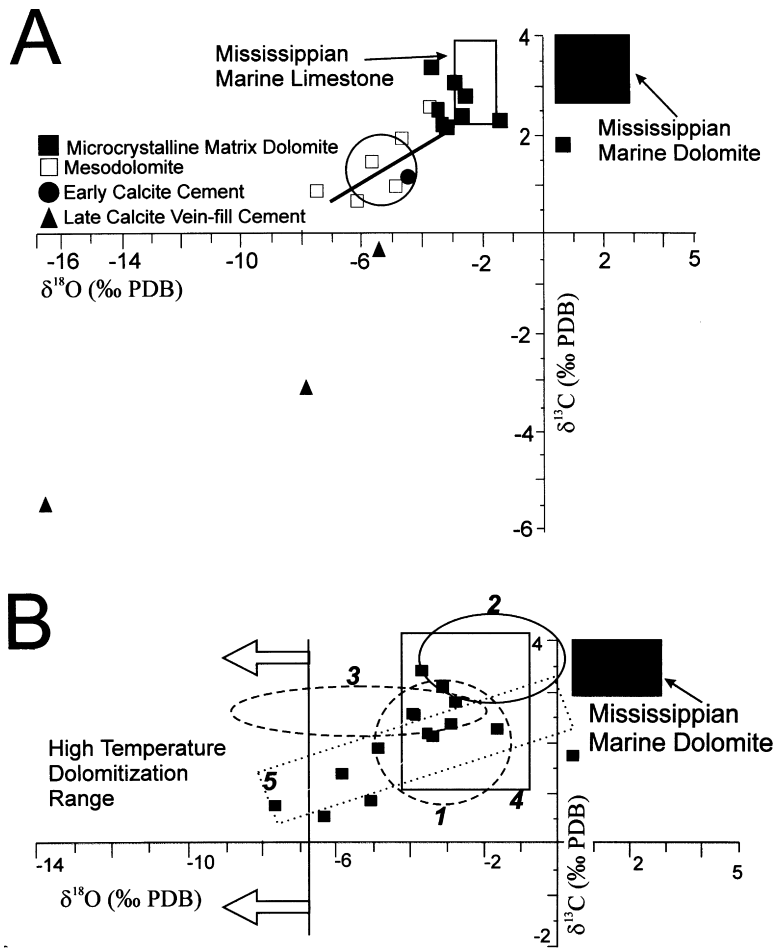


Fig. 6. (a) Stable oxygen and carbon isotopic composition of dolomites and calcites in core from the Moose field. Note the covariant trend in the mesodolomite samples that is indicated by the solid line. The correlation coefficient (R^2) for the six mesodolomite samples is 0.79. (b) Comparison of the Moose field matrix dolomite and mesodolomite stable isotope values (solid squares) to those from other Mississippian dolomites. The regions enclosed by a solid line indicate early matrix dolomite, by a dashed line indicate pervasive matrix dolomite, and by a dotted line indicate a late dolomite phase. Fields are: 1 Quirk Creek field (Turner Valley Formation; Al-Aasm and Lu, 1994); 2 and 3. Blueberry field (Debolt Formation; Durocher and Al-Aasm, 1997); 4 and 5. Pekisko-Sylvan field (Pekisko Formation; Al-Aasm et al., 1996; unpublished data).

4.2.2. Remanence directions

The NRM intensities are between 1.0×10^{-3} and 1.0×10^{-5} A/m with the dolomite specimens in the upper portion and the anhydrite specimens in the lower portion of the range. Almost all of the specimens have very steep, normal-polarity NRM directions (Fig. 10). Upon demagnetization, they tend to exhibit two magnetization components: a low temperature, low coercivity component removed below 240°C or 25 mT, and a high temperature, high coercivity component that is typically removed between 240 and 400°C or 25 and 100 mT. These directions can be seen by examination of Fig. 11, where the horizontal and vertical components of the magnetization are shown separately. In thermally demagnetized specimens, no coherent signal was apparent above 400–450°C because their intensities either fell below the magnetometer noise level or increased sharply from the growth of new magnetic minerals (Fig. 11).

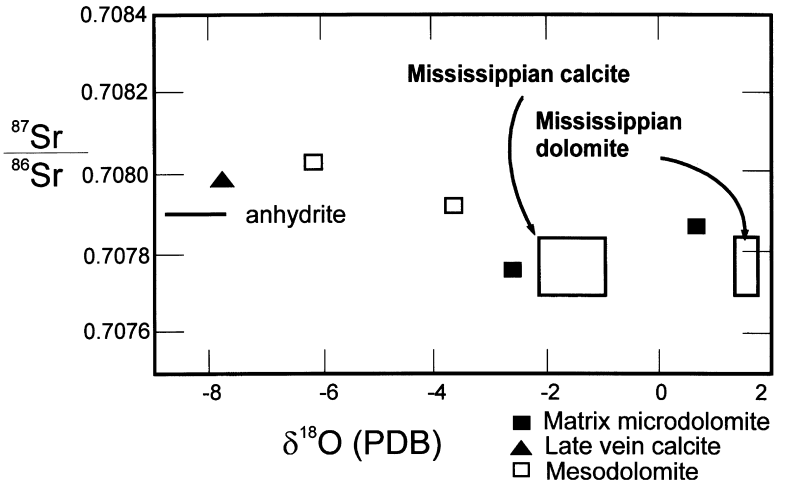


Fig. 7. $\delta^{18}O$ (VPDB) versus $^{87}Sr/^{86}Sr$ compositions for the dolomites and a calcite-vein sample. The strontium isotope value for the anhydrite is noted on the left side of the plot. Reference values from Denison et al. (1994). See text for details.

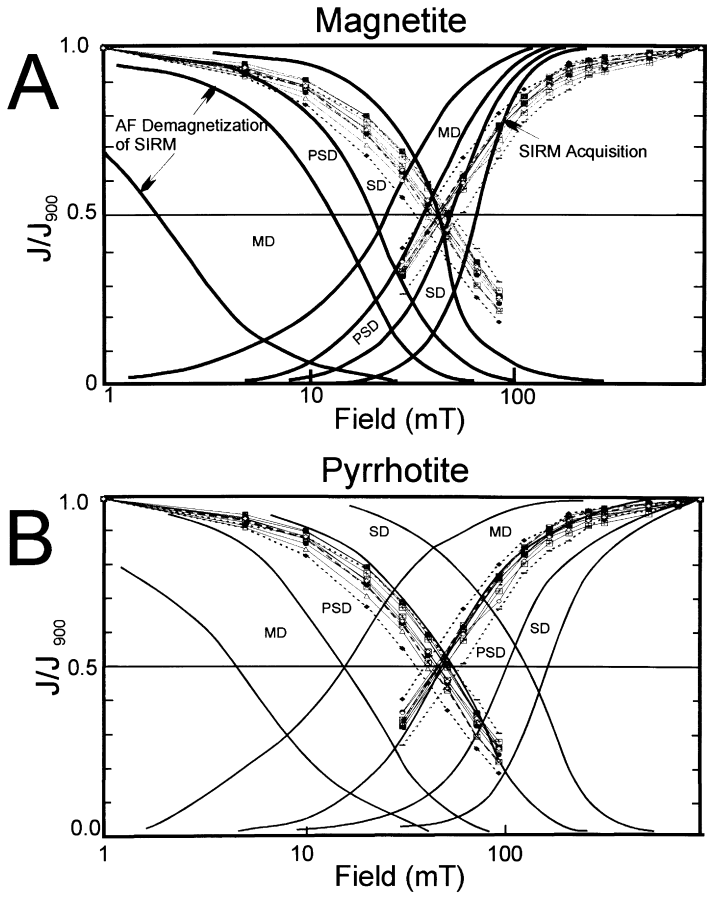


Fig. 8. SIRM crossover plots for: (a) magnetite; and (b) pyrrhotite with IRM acquisition curves and SIRM demagnetization curves for samples from Wells 14 and 23.

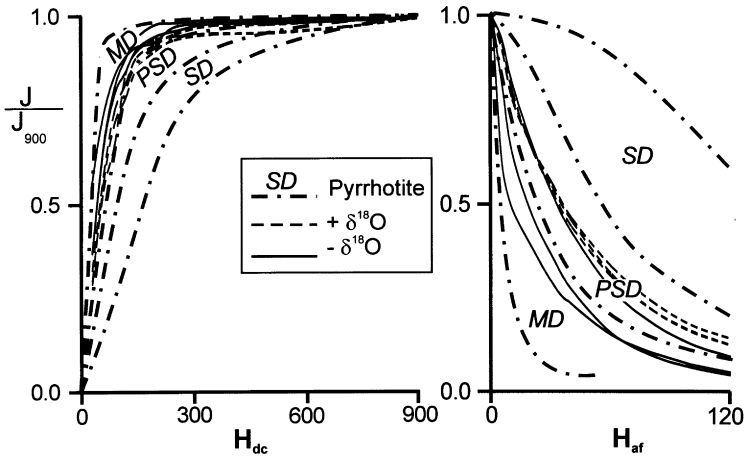


Fig. 9. IRM acquisition and SIRM demagnetization for <2 cm^3 samples showing the covariant trend in the mesodolomite seen in Fig. 6. $+\delta^{18}$O and $-\delta^{18}$O indicate samples that have less-depleted and more-depleted ^{18}O values, respectively. The type curves are for single domain (SD), pseudosingle domain (PSD) and multidomain (MD) pyrrhotite.

The low temperature and coercivity components in Well 14 have a normal polarity (Fig. 12) and are nearly vertical at $D = 322.2°$, $I = 893°$, $\alpha 95 = 5.1°$ on demagnetization (declination, inclination, radius of cone of 95% confidence) (see Table 3). In Well 23, this component is more scattered with an orientation of $D = 92.7°$, $I = 80.3°$, and $\alpha 95 = 14.5°$.

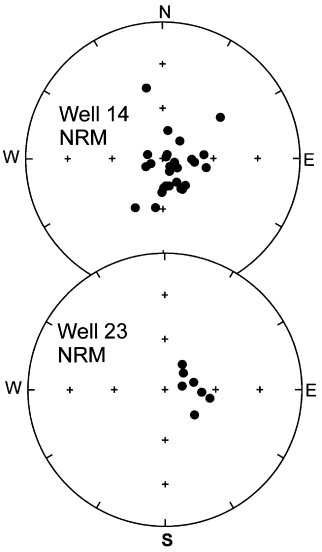


Fig. 10. Equal-area stereogram projections of specimen NRMs from Wells 14 and 23. Solid symbols indicate projection into the lower hemisphere and open circles into the upper hemisphere.

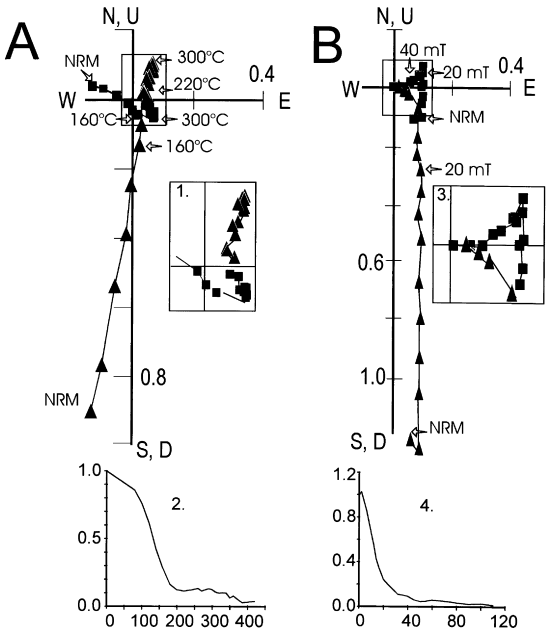


Fig. 11. Demagnetization plots for two samples from Well 14. (a) Thermal demagnetization of specimen 141502. (b) AF demagnetization of specimen 141503. The squares indicate the projection of the magnetization onto the horizontal plane, and horizontal component and triangles the projection onto the vertical component. Axes are north (N), east (E), south (S), west (W), up (U) and down (D). Axial intensities are expressed as a ratio to the natural remanence intensity. Insets 1 and 3 show a blowup of the area near the origin and insets 2 and 4 show the decay in remanence intensity as demagnetization progresses.

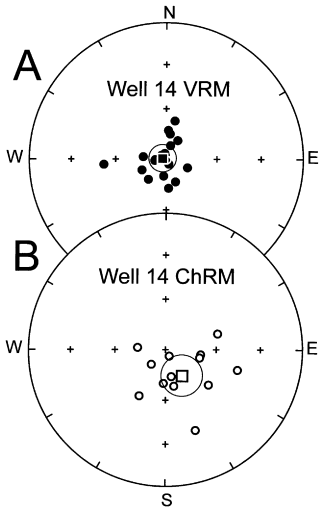


Fig. 12. Equal-area stereogram showing: (a) low-temperature or viscous remanence component from Well 14; (b) high-temperature or characteristic remanence component from Well 14. Solid symbols indicate projection into the lower hemisphere and open circles indicate projection into the upper hemisphere. The mean direction for each well is shown by a square; the circle is the 95% cone of confidence for that mean.

Comparison of these directions to each other and to the present and DEMF directions of $D = \sim 9.2°$, $I = \sim 76.4°$ and $D = 0.0°$, $I = 67.0°$, respectively, indicates that the measured directions from the two wells are significantly different. This suggests that the viscous low temperature and coercivity components observed in these specimens cannot be assumed to have been caused by the Earth's magnetic field and, thus, cannot be used to orient the core segments.

Table 3
Paleomagnetic components (component: the first number is the well number, the second is the demagnetization type (AF demagnetization, TH thermal demagnetization) and the third indicates the specific component (*L* is low temperature/coercivity, *H* is the high temperature/coercivity). The direction is given by its declination (*D*), inclination (*I*), radius of 95% circle of confidence ($\alpha 95$), precision parameter (*k*), and number of specimens averaged (*N*))

Component	*D* (°)	*I* (°)	$\alpha 95$	*k*	*N*
14-TH-L	278.9	87.0	5.9	31.5	20
14-AF-L	85.6	83.1	9.3	43.4	7
23-TH-L	92.7	80.3	14.5	28.9	5
23-AF-L	–	–	–	–	2
14-TH-H	144.6	−71.4	11.9	12.2	13
14-AF-H	136.6	−44.4	43.4	2.4	7
23-TH-H	117.0	−73.3	15.9	24.0	5
23-AF-H	–	–	–	–	2

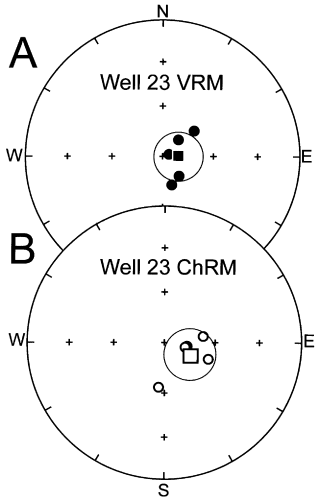


Fig. 13. (a) Low-temperature or viscous remanence component from Well 23. (b) High-temperature or characteristic remanence component from Well 23. See Fig. 12 for symbols.

The characteristic remanence magnetization (ChRM) or high temperature and coercivity magnetization has a reversed polarity in both wells, but is not present in all samples (Fig. 13, Table 3). In Well 14, the ChRM direction calculated from thermally demagnetized samples is $D = 144.6°$, $I = -71.4°$, $\alpha 95 = 11.9°$, whereas the CHRM mean direction from AF demagnetized samples is $D = 136.6°$, $I = -44.4°$ and $\alpha 95 = 43.4°$. Obviously, AF demagnetization does not completely remove the specimens' viscous overprints and, therefore, they never reach a stable endpoint, resulting in poorly defined directions. In Well 23, the high-temperature component mean is $D = 117.0°$, $I = -73.3°$, and $\alpha 95 = 15.9°$. Only two specimens from Well 23 were AF demagnetized, so that a mean was not calculated. The high-coercivity directions for the two specimens are shown in Fig. 13.

4.2.3. Pullaiah-type curves for pyrrhotite

Consideration of the origin of the remanence-carrying magnetic minerals (pyrrhotite and/or magnetite) results in three possibilities. Since no primary depositional remanence is apparent, it could be that: (1) the magnetic minerals deposited or precipitated during sediment deposition have had their magnetic

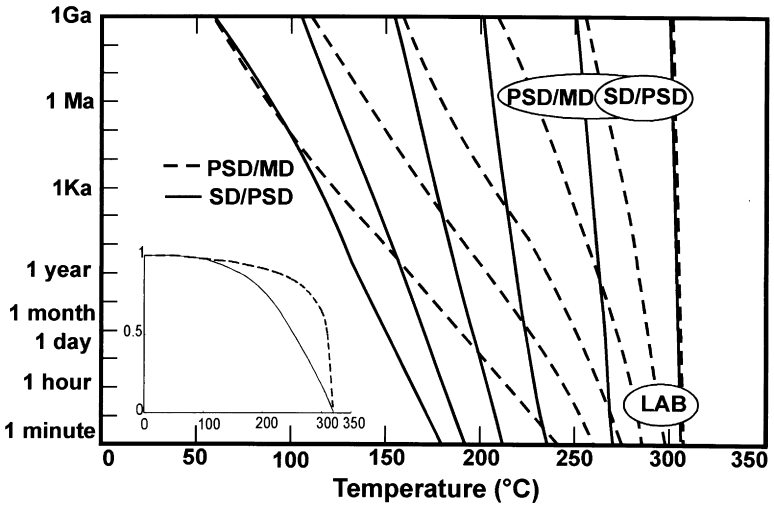


Fig. 14. Time-temperature curves for pyrrhotite, showing the laboratory demagnetization temperatures for times of ~15 min to 1 h that correspond to geological magnetization temperatures and times. The J_s data (inset) was determined from Dekkers' (1989) TTE samples, assuming that the SD/PSD size boundary was ~5 μm and the PSD/MD boundary was ~75–100 μm. The dominant anisotropy was assumed to be magnetocrystalline, based on Dekkers' (1989) calculations.

remanence thermally reset; or (2) the original magnetic minerals have been chemically altered to the extent that their original magnetization has been destroyed and a new magnetization formed; or (3) new remanence-carrying minerals were formed during a chemical event. To examine the first possibility, a temperature/time plot was constructed for pyrrhotite akin to that published for magnetite (Pullaiah et al., 1975). Such curves are used to determine the temperature of formation of a mineral's magnetization for geologic time scales (Ka–Ma) from the temperature at which it is thermally demagnetized at a laboratory time scale (minutes–hours). Because the intensity of saturation magnetization varies with grain-size as well as temperature, both single to pseudosingle domain and pseudosingle to multidomain plots were created for pyrrhotite (Fig. 14). The plots indicate that for laboratory demagnetizing temperatures of 275°C the minimum temperature required for either initial magnetization or remagnetization of pyrrhotite by a thermal event would have been ~225°C for pseudosingle to multidomain pyrrhotite grains and ~260°C for single to pseudosingle domain pyrrhotite grains. Laboratory demagnetizing temperatures of ~310°C require geologic temperatures of ~275–305°C (Fig. 14). If conventional geothermal gradients of ~25°C/km are used, this suggests a minimum burial depth of ~8–9 km would be required to remagnetize pseudosingle to multidomain grains, and ~10–11 km to remagnetize single to pseudosingle domain pyrrhotite grains.

5. Discussion

5.1. Dolomitization

The paragenetic sequence (Fig. 3), determined through petrologic and geochemical analysis, allows constraints to be placed on the timing and origin of dolomitization events. Petrographic evidence suggests that matrix microdolomite formed early in the diagenetic history of the Turner Valley Formation in a marine-dominated system. Its crystal size, nonstoichiometric nature, association with algal filaments and nodular anhydrite, as well as the presence of detrital quartz, suggest formation in a near-shore, peritidal environment (McKenzie, 1981; Patterson and Kinsman, 1982; Muller et al., 1990), whereas grainstone and wackestone facies were deposited in shoals further offshore. This dolomite appears to have formed prior to any appreciable compaction because all stylolites crosscut it. However, most of its isotopic signatures are inconsistent with those of a

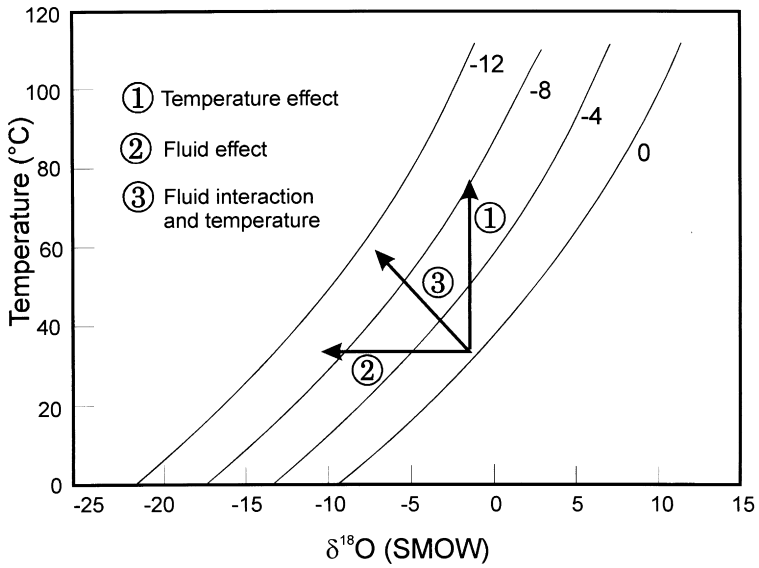


Fig. 15. Oxygen isotope composition of pore water (δ^{18}O‰ SMOW) vs. crystallization temperature (°C) for the Turner Valley matrix dolomite. Isotopic composition of the dolomite is shown as contours (δ^{18}O‰ VPDB). The equation relating temperature to $\delta^{18}O_{water}$ and $\delta^{18}O_{mineral}$ is: $10^3 \ln \alpha_{dol-water} = 3.14 \times 10^6 T^2 - 2.0$ (Land, 1983).

syndepositional and/or very shallow burial origin, which may indicate modifications during successive stages of diagenesis (Malone et al., 1994; Machel, 1997). δ^{18}O isotope values of the matrix dolomite show a negative shift in the magnitude of ~4‰, but the narrow range of δ^{13}C values suggests low water–rock interaction in a rock-buffered system (Fig. 6). The isotopic values of this dolomite also fall very close to those of pervasive dolomite in the Turner Valley Formation from the nearby Quirk Creek Field (Fig. 6b; Al-Aasm and Lu, 1994). $^{87}Sr/^{86}Sr$ ratios of the microdolomite (Fig. 7) fall very closely to the Mississippian seawater value which suggests an unmodified, or only slightly modified, seawater signature. Similar values have been also found for the Quirk Creek microdolomite (Al-Aasm and Lu, 1994).

Recrystallization of microdolomite to mesodolomite (Fig. 5) is supported by petrographic and geochemical evidence. The evidence includes: (1) a gradual increase in dolomite crystal size; (2) the presence of concentric zoning in some dolomite crystals, with irregular boundaries between the core and the rim; (3) a negative shift in oxygen isotopes; and (4) the partial enrichment of radiogenic Sr (Figs. 5–7). Recrystallization of early-formed dolomite often occurs and is thought to take place in shallow, low-temperature and/or deep, high-temperature environments (e.g. Mazzulo, 1992; Smith and Dorobek, 1993; Al-Aasm and Lu, 1994; Kupecz and Land, 1994; Neilsen et al., 1994; Yoo and Lee, 1998). It could alter isotopic and geochemical signatures within the dolomites (Malone et al., 1994; Durocher and Al-Aasm, 1997). In the present study, microdolomite is believed to have recrystallized to mesodolomite during the shallow to intermediate burial stage, post late calcite and silica cementation. The isotopic variations of mesodolomite show a covariant trend between oxygen and carbon isotopes (Fig. 6a). A similar trend in isotopic values has been observed in the Mississippian carbonates from the Pekisko Formation in the Sylvan Lake Field in Alberta (Fig. 6b; Al-Aasm et al., 1996). Al-Aasm and Lu (1994) suggested that the pervasive dolomitization at Quirk Creek Field occurred as the result of mixing of phreatic fresh waters with marine waters or brines, possibly during the Pennsylvanian. The covariant trend can be attributed to evolving porefluids with depth and/or mixing of two types of fluids (cf. Frank and Lohmann, 1995).

Petrographic, burial history and isotope data can be used to interpret changes in the oxygen isotope composition of the diagenetic fluids responsible for

the alteration of early matrix dolomite in the Turner Valley Formation during progressive diagenesis. The relationships among mineral $\delta^{18}O$ values, pore waters $\delta^{18}O$ values and temperature for matrix dolomite and its recrystallized product, mesodolomite, is shown in Fig. 15. Three possible pathways for pore water evolution have been suggested. These pathways reflect: (1) changes in temperature due to increasing burial; (2) changes in fluid chemistry due to fluid–rock interaction; or (3) a combination of increasing temperature and fluid–rock interaction.

In the first scenario (trend 1; Fig. 15), an increase in temperature alone is the key factor for recrystallization of the carbonate components and the formation of later diagenetic mineral phases. In this scenario, the composition of diagenetic fluids of marine parentage cannot change significantly during the burial history, i.e. the system is closed. Al-Aasm and Lu (1994) showed that the maximum burial of the Turner Valley Formation in the Quirk Creek Field during the Mississippian–Early Cretaceous interval did not exceed 1 km. It was only during the Late Cretaceous that burial depths increased, and the maximum burial was ~6 km. If a geothermal gradient of 25°C/km and an initial surface temperature of about 20°C are used, then the maximum burial temperature for the recrystallization of early dolomite is ~45°C whereas temperatures of ~170°C are possible for the formation of later mineral phases (assuming maximum burial of 6 km). However, as mentioned previously, this model would require a completely closed hydrodynamic system, which is a situation difficult to obtain in nature. Thus, this requirement poses difficulty in explaining the diagenetic repartitioning of stable isotopes.

The second scenario assumes that most of the diagenetic changes occurred at lower temperatures due to changes in pore water isotope chemistry from the incursion of meteoric waters following deposition of the Turner Valley Formation (trend 2; Fig. 15). This model may explain the recrystallization of matrix dolomite to mesodolomite. However, this model cannot explain adequately the slight variations in $\delta^{13}C$ isotopes, nor does it explain the changes in oxygen isotopes in the subsequent diagenetic phases. The negative shift in oxygen isotopes is much greater than would result from the incorporation of post-Mississippian (Permian) meteoric fluids alone. The original oxygen isotopic composition of meteoric fluids was estimated to be −5.4‰ SMOW, assuming the paleolatitude of the study area during Mississippian and immediately post-Mississippian times to be approximately 20°N (cf. Durocher and Al-Aasm, 1997). This scenario may have been valid early in the diagenetic history where the possibility of post-Mississippian exposure may have had occurred. However, in the samples investigated in this study, no clear evidence of paleo-exposure surfaces have been found, although earlier studies on Mississippian carbonates in southern Alberta (e.g. Al-Aasm and Lu, 1994) suggested the possibility of post-Mississippian subaerial exposure.

The third scenario involves increasing temperature accompanied by progressive changes in pore–water isotopic chemistry due to water–rock interaction and/or mixing of fluids (trend 3, Fig 15; e.g. Frank and Lohmann, 1995). This scenario explains both the covariation in stable isotopes and the formation of later diagenetic stages during burial. Modification of the original marine pore waters, whether due to incursion of meteoric waters at shallow burial or due to increasing water/rock interactions and temperature, explains both the negative shift in oxygen isotopes and the slight shift in carbon isotopes with the original values in the matrix dolomite.

5.2. Calcite cementation, silicification and anhydritization

Calcite cementation occurred during several intervals. Stable oxygen and carbon isotopes of an early calcite (Fig. 6) both show a slight shift with respect to Mississippian calcite values (Lohmann, 1988). This shift could have resulted from recrystallization of this cement during the shallow burial stage. In contrast, both oxygen and carbon isotopes in the late vein-fill calcite cement have more depleted $\delta^{18}O$ and $\delta^{13}C$ values. These shifts can be explained by increasing temperature during later diagenetic stages and the incorporation of light carbon. Light carbon could originated from thermal abiotic reactions at depth (Machel et al., 1996). The $^{87}Sr/^{86}Sr$ ratio of this late calcite (Fig. 7) is slightly enriched in radiogenic Sr in comparison to Mississippian seawater, and is very comparable to mesodolomite suggesting that the strontium was sourced internally, rather than from

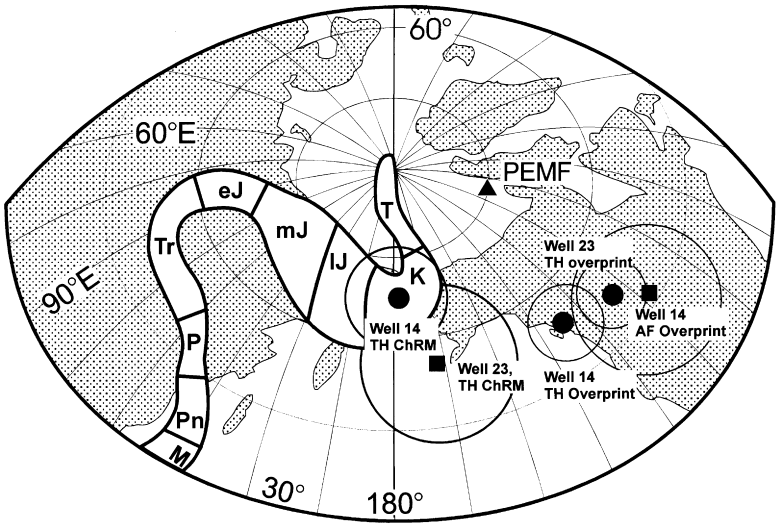


Fig. 16. Apparent polar wander path for the North American craton from Van der Voo (1990, 1993) for the Mississippian through Tertiary. The viscous component and ChRM paleopoles for both wells are shown. A triangle indicates the location of the PEMF.

highly enriched Sr-sources such as shales and/or basement rocks (Mountjoy et al., 1992).

Silica replacement of carbonates is demonstrated by the replacement and preservation of internal skeletal structure of fossils (Fig. 5b). Silicification predates the formation of the mesodolomite. Sponge spicules were observed in zones where preservation of early fabrics occurred, indicating that the silica could originate from other dissolved spicules and thus be biogenic. Alternatively, Al-Aasm and Lu (1994) suggested that the source of silica in the Turner Valley Formation from the Quirk Creek Field could be related to a post-Mississippian disconformity. We noted previously that the late quartz occurs with vug-rimming dolomite and other late diagenetic minerals.

The association of the secondary anhydrite with other late diagenetic phases suggests that these minerals precipitated from basinal brines generated during burial. In contrast, earlier nodular anhydrite may represent primary deposition in a nearshore, peritidal setting, associated with microdolomite. The $^{87}Sr/^{86}Sr$ ratio of the earlier anhydrite falls close to the Mississippian seawater composition (Fig. 7).

5.3. Magnetizations

The normal-polarity paleopoles from the viscous remanence of both wells are significantly different from the PEMF and DEMF poles (Fig. 16), suggesting a drilling-induced remanence because it is almost parallel to the coring direction. Shi and Tarling (1997) have suggested that a drilling-induced remanence would be formed in physically metastable minerals such as pyrrhotite by drilling vibration. Given that pyrrhotite is interpreted to be a major carrier of magnetization in our rocks, it is possible that such a remanence is present. However, one might still expect to see a PEMF or DEMF signal in the demagnetization spectrum of the specimens. One limit is that the formation temperatures were ~100°C, so that any magnetization due to the PEMF or DEMF would have to be removed at temperatures >100°C but prior to removal of the ChRM. In this reservoir, the ChRM is reversed in polarity and, as the Zijderveld diagrams indicate, starts to be removed at relatively low temperatures so that isolation of a PEMF or DEMF signal in the viscous component is unlikely. In effect, the combination of a strong DIRM and a reversed ChRM appears to swamp any PEMF or DEMF signal that might be present.

A single ChRM is present in the two wells that is directed southeast and up. This direction is more clearly defined in Well 14, mostly because of the limited number of specimens representing Well 23. The ChRM paleopole for Well 14 falls on the

Cretaceous section of the APWP (Fig. 16). Although the ChRM paleopole for Well 23 is slightly offset from the APWP, it is based on only five samples with a large $\alpha 95$ that overlaps the APWP. This paleopole is obviously not recording a primary magnetization because a Mississippian paleopole would have a ChRM direction of about $D = 303°$, $I = 20°$. The invariable presence of the Cretaceous component and the lack of any Mississippian magnetizations indicate that these rocks have been pervasively remagnetized during the Cretaceous, either by a thermal event or by chemical alteration due to water/rock interactions. Of the two options, the latter is more plausible because the "Pullaiah-type" pyrrhotite curves show that the temperatures required for thermal resetting of the pyrrhotite are greater than the postulated burial curves for the nearby Quirk Creek Field possibly suggested (Al-Aasm and Lu, 1994). Additionally, if magnetite is present, the required temperature for thermal resetting would be >200° higher. There is no evidence for greenschist-facies metamorphic temperatures in the cores. However, a relationship between orogenic activity, hydrocarbon accumulation, fluid flow and chemical remagnetization is possible (Oliver, 1986, 1992; McCabe and Elmore, 1989). A similar magnetization direction was seen by Lewchuk et al. (1998) in the Mount Head Formation at the Shell Waterton Field to the south of the Moose Mountain area. They concluded that the direction resulted from fluid–rock interaction during the Laramide Orogeny. Enkin et al. (1997) sampled the upper Livingstone Formation at the surface in the Moose Mountain area. They concluded that the dominant ChRM was a predeformational chemical remagnetization that was formed during the Laramide Orogeny, no earlier than 75 Ma. Thus, the presence of a Cretaceous remagnetization is not unexpected in these cores.

5.4. Paleomagnetism, isotopes and fluid flow

The paleomagnetic data give a definite Cretaceous age with no indication of any Mississippian directions. This is most straightforwardly interpreted to indicate that the rock underwent major alteration during the Cretaceous because the timing coincides with the Laramide orogeny and the simplest hypothesis would be that an orogenically induced remagnetization formed during a massive fluid flow event, as discussed in the introduction. However, the isotopic evidence for a massive Cretaceous fluid flow event is not present: i.e. the Sr isotopic ratios are not significantly enriched, suggesting that extrabasinal fluids were not important in the diagenetic evolution of pore fluids. Additionally, the covariant isotopic trend of dolomites and their isotopic values do not extend into the high temperature dolomitization range (Fig. 6b). The question that then arises is: what mechanism (s) could totally reset the paleomagnetic signal without totally resetting the isotopic signal?

One possible solution is suggested by the IRM acquisition and demagnetization data for samples along the mesodolomite covariant trend. They show an increasing magnetic grain size with decreasing $\delta^{18}O$ values. This increase in magnetic grain size suggests that the magnetization is a chemical remanent magnetization that is 'locked in' during crystal formation as the crystals precipitated from solution. The correlation of magnetic grain size and isotopic values suggests that recrystallization of microdolomite to mesodolomite may be responsible for the chemical remanent magnetization observed. Recrystallization of dolomite occurs as a result of dissolution and reprecipitation of the initial material. During this process, dissolution and reprecipitation of the magnetic minerals could occur as well, thus realigning the ChRM (Housen et al., 1993). The paleomagnetic age determined would then indicate that this recrystallization occurred during the Cretaceous.

Secondly, the magnetization could have formed earlier during the formation of the matrix dolomite. However, the change in grain size of the magnetic mineral, as apparent in the IRM acquisition, is associated with the mesodolomite phase. Many of the matrix dolomite stable isotopic values are close to the postulated Mississippian values, suggesting a Mississippian origin, whereas the dominant ChRM is Cretaceous, demonstrating the age paradox mentioned earlier. Thus, we consider it less likely that the matrix dolomitization process formed this chemical magnetization.

As a third possibility, we noted that bitumen occurs in mesodolomite cores (Fig. 5c). It is possible that fluid mixing due to the influx of hydrocarbon-bearing

fluid, as suggested by the model proposing changes in pore water chemistry, caused a resetting of the magnetization. However, the correlation between effective magnetic grain size and stable isotope values in the mesodolomite is more difficult to explain because bitumen replacement would have to occur preferentially in the larger mesodolomite crystals in order to produce the relationship seen.

The preferred explanation that recrystallization occurred during the Cretaceous apparently conflicts with our earlier suggestion that recrystallization occurred during the shallow to intermediate burial stage. However, the burial curve constructed for the Quirk Creek Field, just east of the Moose Field suggests a long interval of shallow to intermediate burial during the late Paleozoic and most of the Mesozoic (Al-Aasm and Lu, 1994) for this area. Significant burial, resulting from Laramide orogenesis did not occur until the Early Cretaceous. The absolute timing of the formation of mesodolomite is therefore difficult to constrain. Additionally, the presence of a Cretaceous chemical magnetization does not preclude earlier recrystallization events from having occurred; it simply suggests that the CRM would date the last recrystallization event in the dolomite and it was most probably associated with the Laramide Orogeny.

Thus, we interpret the combined petrographic, paleomagnetic, rock magnetic and geochemical evidence to indicate that recrystallization of early-formed microdolomite and the formation of a chemical remanent magnetization occurred during the Cretaceous, most probably coincident with the Laramide Orogeny. It therefore follows from this interpretation that a significant orogenically induced fluid flow, due to the Laramide orogeny, did not occur through the Turner Valley Formation in the Moose reservoir. However, we emphasize that this conclusion cannot be extended to other reservoirs, as the evidence on which it is based is unique to this formation.

6. Conclusions

1. It is apparent from this and other concurrent studies that paleomagnetic and rock magnetic techniques are highly sensitive to even relatively minor events recorded in the diagenetic mineral sequence. As shown in this study, it may be possible to constrain the timing of diagenetic events but, unless it is possible to completely isolate a component for both paleomagnetic and geochemical sampling, it may be very difficult to absolutely date a particular event. Nevertheless, paleomagnetism does provide the potential to absolutely date diagenetic events that cannot be readily dated by other methods.
2. In the Moose Field, the viscous remanence directions are not useful for orienting the core, because this component appears to have been induced during drilling. This may be because pyrrhotite is the main remanence carrier and experimental work by others has suggested that it may be susceptible to shock vibrations such as drilling might induce. There is no indication of any primary Mississippian magnetization directions in either well used in this study. The ChRM isolated in both wells is a secondary chemical magnetization that was formed during the Cretaceous.
3. Petrographic and geochemical analysis suggests a complex diagenetic sequence involving episodes of silicification, dolomite formation, brecciation, anhydritization, compaction, and late fluorite and calcite vein-filling. The analysis indicates the presence of two major followed by two minor dolomite phases: early matrix dolomite, mesodolomite, pseudomorphic dolomite, and vug-rimming dolomite. The early matrix dolomite has $\delta^{18}O$, $\delta^{13}C$, ^{87}Sr values that are only slightly altered from postulated Mississippian dolomite values. In contrast, the mesodolomite shows a covariant trend in the isotopic values that suggests increasing temperature accompanied by progressive changes in pore water isotopic chemistry caused by water–rock interaction and/or mixing of fluids. Petrographic evidence suggests that the mesodolomite was formed by the recrystallization of the matrix dolomite during shallow to intermediate burial.
4. The rock magnetic analysis indicates that the Cretaceous ChRM is associated with the mesodolomite phase. The apparent age contradiction between the isotope (Mississippian) and paleomagnetic (Cretaceous) data can be resolved in two ways. If the magnetization is held by the mesodolomite, the ChRM dates its formation by recrystallization from the matrix dolomite, and the timing

is constrained by the paleopole to have occurred during the Laramide Orogeny. This possibility suggests that limited extrabasinal fluid inflow occurred in this area. Alternatively, the magnetization may have resulted from the replacement of the mesodolomite by bitumen, in which case we are dating the episode of hydrocarbon-bearing fluid migration during the Cretaceous.

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